

The Agricultural Data Arms Race:



Exploiting a Tractor Load
of Vulnerabilities In The
Global Food Supply Chain.

(in good faith)

SICK.CODES

A man, "Using a
hand tractor for
cultivation.
Falls City
Farmsteads,
Nebraska"

Rothstein,
Arthur, 1915-
1985,
photographer
(Public Domain)

9



John Deere 7450
ProDrive Forage
Harvester



Disclaimer

- None of research was paid for
 - All research was done in good faith
 - Nothing today represents our employers, past employers, or future employers
 - None of us are under gag orders*
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-



Who am I?

Sick Codes – good hackerman

<https://github.com/sickcodes>

<https://twitter.com/sickcodes>

<https://linkedin.com/in/sickcodes>

<https://sick.codes>



Docker-OSX

Run Mac in a Docker! Run near native OSX-KVM in Docker!
Forwarding! CI/CD for OS X!

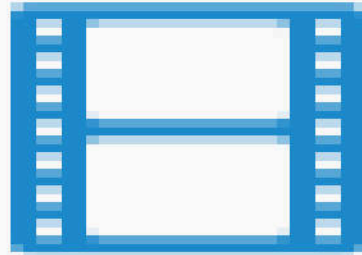
 Shell  14.9k  697

osx-serial-generator

Mac Serial Generator - Generate complete sets of Serial Numbers
for OSX-KVM, Docker-OSX and of course, OpenCore.

 Shell  1.2k  57

Why is this important?



Scenarios

- "The sprayers are programmed by the hacker to unevenly spray the chemicals on the crops, applying ten times the chemical on certain parts of the field, and a tenth the dose on other parts."

Skewed risks

- Denial of Service BIG impact
- "hacker uploads a firmware update that inserts an offset into the GPS locations used by the target."

Real threats

- "offset of (say) 100 yards North of where it should be. The target navigates itself onto a highway, into a river, through a fence, over a cliff, or whatever. Target is destroyed,"

Conor Cagney:

<https://sick.codes/leaky-john-deere-apis-serious-food-supply-chain-vulnerabilities-discovered-by-sick-codes-kevin-kenney-willie-cade/#comment-41658>

More Real threats

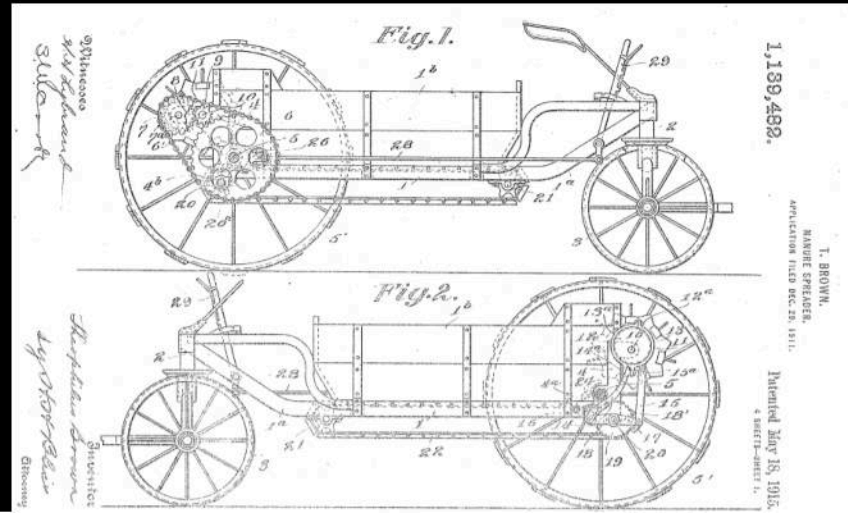
- "The sprayers are programmed by the hacker to unevenly spray the chemicals on the crops, applying ten times the chemical on certain parts of the field, and a tenth the dose on other parts."

Why did we even look at Ag?

Nobody else was.

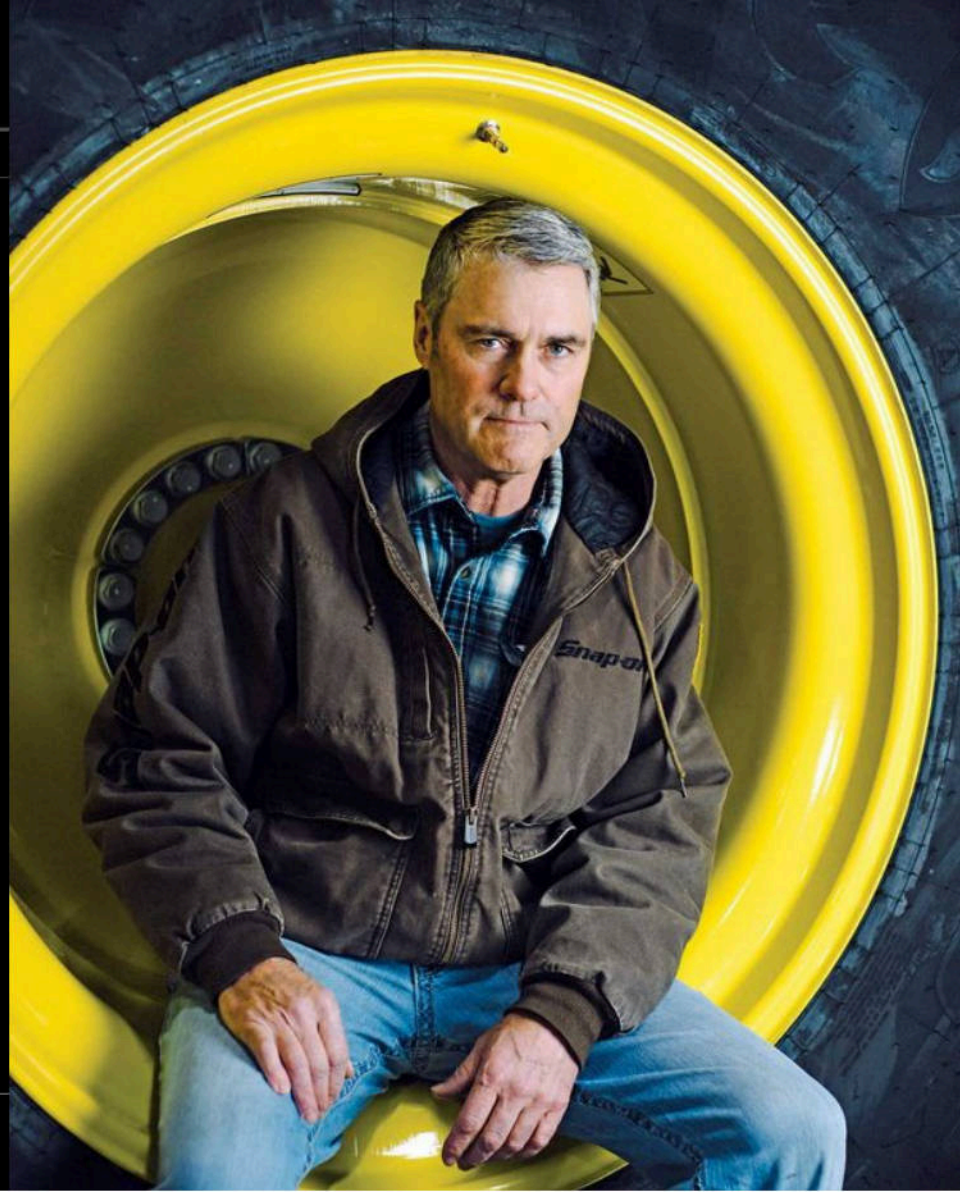
Willie Cade

Grandson of
Theo Brown who
was spent 30
years on John
Deere Board.



Kevin Kenney

- Nebraska
 - Engineer
 - Farmer
-
- Photographer: Walker Pickering for Bloomberg Businessweek



The hackers

- Sick Codes <https://twitter.com/sickcodes>
- wabaf3t <https://twitter.com/wabafet1>
- D0rkerDevil <https://twitter.com/D0rkerDevil>
- johnjhacking <https://twitter.com/johnjhacking>
- rej_ex https://twitter.com/rej_ex
- w0rmer <https://twitter.com/0x686967>
- ChiefCoolArrow <https://twitter.com/ChiefCoolArrow>
- Kelly <https://twitter.com/kaoudis>

Precision Agriculture

- Every single farm is **connected**.
- 5G, LTE, 2G/3G, LoRa, WiFi, GPS, GPRS, WAAS, RTK, NTRIP

Precision Ag Accuracy

GPS

GPS+RTK



Accuracy

- <https://www.flickr.com/photos/hindrik/50411142851/>

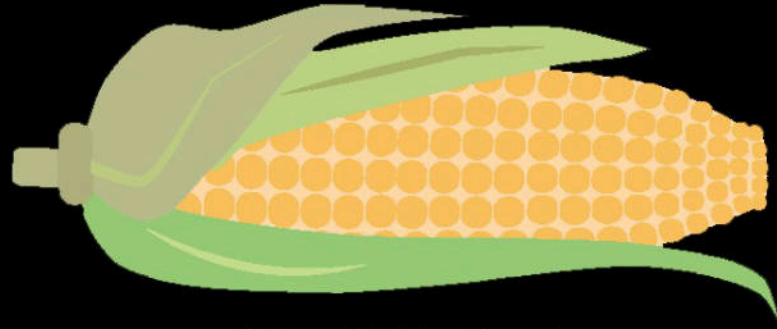


Accuracy

SF1

SF2

RTK



12 INCHES

What does
the data
do?



"Trade Secrets"

- "Does sharing data with data analytic providers destroy its secrecy?"
- "{I think the answer is probably "no," provided sharing is done anonymously."
- <https://www.aglaw.us/janzenaglaw/2015/9/30/is-farm-data-a-trade-secret>



Biofuel



Other data usage

- Carbon credits (carbon offset market)
- Mandatory
- Voluntary
- *Shannon Sedgwick – Farmer turned Managing Director*

[Settings](#)[Help](#)[About](#)[Troubleshooting](#)[Last Update: June 30, 2021](#)[Logout](#)

What would you like to simulate?

[Start Simulation](#)

Machine

Self-Propelled Forage Harvester



Product models and options may not be available in all regions.

[Expand all](#)

Display

GreenStar™ 3 CommandCenter™



Machine

Self-Propelled Forage Harvester



Internet of...





JOHN DEERE

Operations Center



Preference Navigation



General >

Remote Display Access >

Send File >

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Desktop | Contact Us

Service Admin Portal

JD Quote

Configurator

My John Deere

Start

Agreement

Mana

Deere Master Template

Dealer Template

Template : Remote Support Pa

Dipping my feet in...



Sign In

Next

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MOTHERBOARD

TECH BY VICE

Bugs Allowed Hackers to Dox John Deere Tractor Owners

A security researcher found two bugs that allowed him to find customers who had purchased John Deere tractors or equipment.



By [Lorenzo Franceschi-Bicchierai](#)

"All" tractors

Correction, April 22, 1:22 p.m. ET: This article and headline have been corrected to clarify that not "all" owners of John Deere equipment were potentially affected by these vulnerabilities.



wabafet

@wabafet1



wabafet @wabafet1

I'm a passionate self taught researcher that likes to bug bounty hunt and program

295 Following **200** Followers

 Joined April 2021



hey bro how do i get ahold of vdp for deere
got mad shit on them like 5 xss so far i can
only imagine what the rest my toolkits gonan
find in regards to p2 leaks

Apr 23, 2021, 1:59 PM

You accepted the request

Haha holy shit

You got signal?

Apr 23, 2021, 2:40 PM ✓



sickcodes 4:44 PM

im getting H1 or JD to comment
like

fuck doing this for free

one sec sending audio



wabaf3t 👑 4:45 PM

k

well at the end of the day id rather do it for free than loose food

someones got to save their dumbasses

EDITED

John Deere  @JohnDeere
Nothing runs like a Deere
222 Following 196.2K Followers
 Joined January 2009

have a few pretty serious security bugs to report

1:38 AM ✓

i can be reached at wabafet@tutanota.com

2:12 AM ✓

We encourage the ethical hacker community to report any possible vulnerability that is identified in our assets as we truly appreciate their work towards securing John Deere. Please submit vulnerabilities to us here: bit.ly/32LHxxO. We ensure safe harbor when vulnerabilities are reported in a good faith. At this time, do not have a bug bounty program or offer any monetary rewards for these submissions.



2:18 PM

ok i sent them in thank u for the reply if i may suggest can u add multiple bugs allowed or something i had to submit u like 5 bugs in 1 secription field with no way to comment

2:24 PM ✓

ations

Help

Contact Us

The following errors must be corrected before you

"Purchase Order Number " is not valid

Purchase Order Number

ipt>alert(1)</sc

John Deere Unit that
issued the Purchase Order

Select... ▼

Email Address

Create Invoice

1

OK

** all fields a

Supply Network

Hi, Guest ▾

Devl / Qual **Supply Network**

Hi, Guest ▾

[Create e-Invoice](#)[View e-Invoice Status](#)[Electronic Funds Transfer \(EFT\)](#)[Remit To Address Change](#)

Email Address

Supplier Number

Invoice Number

** all fields are required

[Search](#)[Reset](#)

New Functionality!

Now, you just have to provide a PO number and E-Invoice will filter and display the units related to who wrote the PO.

[Got it!](#)



JOHN DEERE

Supply Network

Hi, Guest ▾



e-Invoice

Applications

Help

Contact Us

Create e-Invoice

View e-Invoice Status

Electronic Funds Transfer
(EFT)

Remit To Address Change

INV0106 - No invoices found.

** all fields are required

Email Address

*@deere.com

Supplier Number

*

Invoice Number

0

Search

Reset

Loading





JOHN DEERE

Supply Network

Hi, Guest ▾



e-Invoice

Applications

Help

Contact Us

Create e-Invoice

View e-Invoice Status

Electronic Funds Transfer
(EFT)

Remit To Address Change

INV0106 - No invoices found.

****** all fields are required

Email Address

*@deere.com

Supplier Number

*

Invoice Number

0

Search

Reset

We're sorry, there was an application error.

Please [contact the Help Desk](#) by phone or email with the Diagnostic Information below.

You may want to print this page as it has important information about the error.

[Go back to the page I was viewing](#)

Diagnostic Information

User ID

Server Time

2021-04-25 20:10:33.881

Application Name

e-Invoice

Application URL

/invoice/servlet/com.deere.u90242.invoice.view.servlets.SelectPOServlet

Error Message

Offending query is: SELECT COUNT(PO_NUM) TOTAL FROM
(SELECT DISTINCT PO_NUM FROM
DJDBP01.PURCH_HRZN_PO_DTL WHERE PO_NUM = " UNION
ALL SELECT DISTINCT PO_NUM FROM
DJDBP01.SM_IND_ORD_HDR WHERE PO_NUM = ")

Sign in to Access My JDSN

Already Have a User ID?

[Sign In](#)

No User ID?

[How to Obtain Access?](#)

JDSN Applications

[Applications by Name](#)

[How Do I...](#)

Filter Results:

Application Name	Functional Area
Achieving Excellence	Supplier Quality & Performance
Change Request	EPDP/SCI/Collab
Compliance Information Collection (CIC)	Compliance
Credit Memo	Order Management
Delivery Receipts	Order Management
Demand Report	Order Management
e-Invoice	Order Management
EOD Commercial Invoice	Global Trade Management
EOD Export ESO System	Global Trade Management
EOD Hazardous Material Form	Global Trade Management
EOD Ship Notice	Global Trade Management
Forgot My Password	Other
Fulfill Email Preferences	Order Management
Indirect Open Orders	Order Management
Invoice and Payment Status	Order Management
Invoice Attachments	Order Management
JD Color Standards	Other
JD CROP	Cost Management
JD Supplier Warranty	Supplier Quality & Performance
JDSN Collaboration	EPDP/SCI/Collab
Manufacturing Critical-path Time (MCT)	Order Management
Non-Conformance Corrective Action (NCCA)	Other
North American Freight Invoices	Logistics
Open Triggers	Order Management
Shipment Tracking	Logistics
Standard Shipping Label	Global Trade Management
Supplier Capability Planning	Order Management
Supplier Contacts	Supplier Information
Supplier Diversity	Other
Supplier Education Program	Other



JOHN DEERE

LMSSSO

Welcome Sick Codes | [Logout](#)

Application Error

Message: java.io.FileNotFoundException: SRVE0190E: File not found: /P/

null

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Data Privacy Consent Statement

Talent Central Data Use Guidelines

John Deere Employees

By accessing Talent Central (including John Deere University), I acknowledge that I have read and understand the [Talent Central Data Use Guidelines](#).

Other users accessing John Deere University

Accept

Decline



Equipment Reservation

 All Fields Required, Email must be valid

Email for Reservation (name@johndeere.com) *

Start Date & Time *

April 23, 2021 8:00AM 

End Date & Time *

April 24, 2021 5:00PM 

Retrieve Available Equipment

Showing 10 entries

Filter entries 

Name	Type	PIN	Model Year	Availability
No data available in table				

All Equipment

Showing 100 ▾

Name

8RX 370 (F9a)

9RX 490 (F11)

9RT 570 (F11)

410R

DB60 36R20 (Gaines)

612R (D8)

Self-Propelled Sprayer

Showing 101 to 106 of 106 entries

[Previous](#) [1](#) [2](#) [Next](#)

Equipment Details



DB60 [REDACTED] PIN: [REDACTED]

Equipment Type: Implement - Planter

Model Year: MY22

Additional Notes: [REDACTED]

Requires additional equipment

Upcoming Schedule:

Reserved By	Start	End
[REDACTED] johndeere.com	8:00am on April 27th	5:00pm on April 28th

19 [20:58:04] [INFO] fetching columns for table 'reservations' in database 'machinebook'

20 [20:58:05] [INFO] fetching entries for table 'reservations' in database 'machinebook'

21 Database: machinebook

22 Table: reservations

23 [810 entries]

	id	machine_id	reserve_id	implement_id	name	end	start
--	----	------------	------------	--------------	------	-----	-------

27 [20:58:07] [WARNING] console output will be trimmed to last 256 rows due to large table size

28	618	12		NULL	ndeere.com	2021-03-19 17:00:00	2021-0
29	619	114		NULL	m@johndeere.com	2021-03-18 17:00:00	2021-0
30	620	7		NULL	hndeere.com	2021-03-19 12:00:00	2021-0
31	621	114		NULL	m@johndeere.com	2021-03-19 13:00:00	2021-0
32	622	21		NULL	ndeere.com	2021-03-19 12:00:00	2021-0
33	623	88		NULL	m@johndeere.com	2021-03-19 17:00:00	2021-0
34	624	27		NULL	ndeere.com	2021-03-19 15:00:00	2021-0
35	625	99		NULL	@johndeere.com	2021-03-26 16:00:00	2021-0
36	626	6		NULL	m@johndeere.com	2021-03-22 17:00:00	2021-0
37	627	30		49	ndeere.com	2021-03-22 12:00:00	2021-0
38	628	12		NULL	ndeere.com	2021-03-22 17:00:00	2021-0
39	629	90		NULL	deere.com	2021-03-23 17:00:00	2021-0
40	630	112		50	hndeere.com	2021-03-22 17:00:00	2021-0
41	631	104		NULL	deere.com	2021-03-23 17:00:00	2021-0
42	632	104		NULL	deere.com	2021-03-22 17:00:00	2021-0
43	633	111		NULL	hndeere.com	2021-03-23 17:00:00	2021-0
44	634	86		NULL	@JohnDeere.com	2021-03-25 17:00:00	2021-0
45	635	21		54	ohnDeere.com	2021-03-22 17:00:00	2021-0

```
1 *****
2 *           Edge Server Files           *
3 *****
```

```
4
5 This directory (where this ReadMe.txt file is located) and its
  subdirectories contain
6 PRPC static content files generated for the following Rule
  Application instance:
```

```
7
8 Rule Application Name: DeereSSOSAML
9 Rule Application Version: 01.01.01
```

```
10
11 For more details consult PDN and look for KB article entitled :
   "How to deploy static content to an Edge Server".|
```

rej_ex & john jackson's CVE

[CVE List▼](#)[CNAs▼](#)[WGs▼](#)[Board▼](#)[About▼](#)[News & Blog▼](#)**NVD**

Go to for:

[CVSS Scores](#)[CPE Info](#)[Search CVE List](#)[Downloads](#)[Data Feeds](#)[Update a CVE Record](#)[Request CVE IDs](#)TOTAL CVE Records: 156599

HOME > CVE > CVE-2021-27653

[Printer-Friendly View](#)**CVE-ID****CVE-2021-27653**[Learn more at National Vulnerability Database \(NVD\)](#)

• CVSS Severity Rating • Fix Information • Vulnerable Software Versions • SCAP Mappings • CPE Information

Description

Misconfiguration of the Pega Chat Access Group portal in Pega platform 7.4.0 - 8.5.x could lead to unintended data exposure.

References**Note:** [References](#) are provided for the convenience of the reader to help distinguish between vulnerabilities. The list is not intended to be complete.

rej_ex & john jackson's CVE

- <https://robertwillishacking.com/cve-2021-27653-march-2021/>



NIST: NVD

Base Score: 4.9 MEDIUM

Vector: CVSS:3.1/AV:N/AC:L/PR:H/UI:N/S:U/C:H/I:N/A:N



CNA:

Pegasystems Inc.

Base Score: 6.6 MEDIUM

Vector: CVSS:3.1/AV:N/AC:H/PR:H/UI:N/S:U/C:H/I:H/A:H

Extract EdgeServer Files

Select Rule Application

DeereKM



Application Version

01.01.01



File Name

Create Zip File

Close

Extract EdgeServer Files

Select Rule Application

DeereKM

Application Version

DeereKM

File Name

DeereKMTranslation

DeereKM_backup

DeereSSO

DeereSSOSAML

Create Zip File

DMSample

KMCommunity

PegaAESRemote

PegaBC

PegaDM

PegaKM

PegaNLP

PegaRULES

PegaUVU

RequestApproval

ServiceRequest

StockTrader

TaskTracker

Theme-Clarity

Theme-Cosmos

```
1 *****
2 *           Edge Server Files           *
3 *****
```

4

5 This directory (where this ReadMe.txt file is located) and its
subdirectories contain
6 PRPC static content files generated for the following Rule
Application instance:

7

8 Rule Application Name: DeereSSOSAML
9 Rule Application Version: 01.01.01

10

11 For more details consult PDN and look for KB article entitled :
"How to deploy static content to an Edge Server".|

```
▼<pagedata>
  <pxUpdateSystemID>[REDACTED]</pxUpdateSystemID>
  <pyClassName/>
  <pxUpdateDateTime>20141201T091133.492 GMT</pxUpdateDateTime>
  <pxFrom>PortalServer</pxFrom>
  <pxMoveImportOpName>[REDACTED]</pxMoveImportOpName>
  <pxUpdateOpName>Administrator</pxUpdateOpName>
  <pyWindowTitle/>
  <pxUpdateOperator>Administrator@pega.com</pxUpdateOperator>
  <pyHasCustomFields/>
  <pyUsage/>
  <pxMoveFromSystem/>
  <pxCreateDateTime>20110323T053818.659 GMT</pxCreateDateTime>
  <pyRuleSet/>
  <pyDescription/>
  <pyPassword>[REDACTED]</pyPassword>
  <pyHistoryObject/>
  <pyTemplateDataField/>
  <pxMoveImportOpId>[REDACTED]</pxMoveImportOpId>
  <pxInsName>PORTALSERVER</pxInsName>
  <pxSaveDateTime>20161227T041629.496 GMT</pxSaveDateTime>
  <pyFormPost/>
  <pyIsTagged>true</pyIsTagged>
  <pyAccessGroup>PRPC:PortalUsers</pyAccessGroup>
  <pyTemplateDefault/>
  <pyTemporaryObject/>
  <pzInsKey>DATA-ADMIN-APPID PORTALSERVER</pzInsKey>
  <pyTemplate/>
  <pyTimeDifferenceShadow>[REDACTED]</pyTimeDifferenceShadow>
  <pxMoveImportDateTime>20121218T134807.752 GMT</pxMoveImportDateTime>
  <pxObjClass>Data-Admin-AppID</pxObjClass>
  <pxTimeDifference>[REDACTED]</pxTimeDifference>
  <pxCreateOperator>Administrator@[REDACTED].com</pxCreateOperator>
  <pyTemplateInputBox/>
  <pySPRuleSetName/>
  <pyReloadForm>true</pyReloadForm>
  <pyDeletedObject/>
  <pxCreateSystemID>pega</pxCreateSystemID>
  <pxLimitedAccess>Dev</pxLimitedAccess>
  <pxCommitDateTime>20180214T153521.000 GMT</pxCommitDateTime>
  <pyLabel>PortalServer</pyLabel>
  <pzOriginalInstanceKey>DATA-ADMIN-APPID PORTALSERVER</pzOriginalInstanceKey>
  <pxCreateOpName>Administrator</pxCreateOpName>
  <pyRuleSetName>Pega-ProcessCommander</pyRuleSetName>
  <pzStatus>valid</pzStatus>
  <pxWarnings REPEATINGTYPE="PageList"/>
</pagedata>
```

```
<pxUpdateOpName>Administrator</pxUpdateOpName>
<pyWindowTitle/>
<pxUpdateOperator>Administrator@pega.com</pxUpdateOperator>
<pyHasCustomFields/>
<pyUsage/>
<pxMoveFromSystem/>
<pxCreateDateTime>20110323T053818.659 GMT</pxCreateDateTime>
<pyRuleSet/>
<pyDescription/>
<pyPassword>[REDACTED]</pyPassword>
<pyHistoryObject/>
<pyTemplateDataField/>
<pxMoveImportOpId>[REDACTED]</pxMoveImportOpId>
<pxInsName>PORTALSERVER</pxInsName>
<pxSaveDateTime>20161227T041629.496 GMT</pxSaveDateTime>
<pyFormPost/>
<pyIsTagged>true</pyIsTagged>
<pyAccessGroup>PRPC:PortalUsers</pyAccessGroup>
<pyTemplateDefault/>
<pyTemporaryObject/>
<pzInsKey>DATA-ADMIN-APPID PORTALSERVER</pzInsKey>
<pyTemplate/>
<pyTimeDifferenceShadow>[REDACTED]</pyTimeDifferenceShadow>
<pxMoveImportDateTime>20121218T134807.752 GMT</pxMoveImportDateTime>
<pxObjClass>Data-Admin-AppID</pxObjClass>
<pxTimeDifference>[REDACTED]</pxTimeDifference>
<pxCreateOperator>Administrator@[REDACTED].com</pxCreateOperator>
<pyTemplateInputBox/>
```


Security Audit Log

Filtered by: .pxCreateDateTime

Displaying 50 records

Create Date/Time ↓	Remote IP Address	Remote Host	User Id	Message	Browser(User-Agent)
4/23/2021 11:03 PM	[REDACTED]	[REDACTED]	[REDACTED]	The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64) AppleW
4/23/2021 11:02 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64) AppleW
4/23/2021 10:56 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64) AppleW
4/23/2021 10:55 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64) AppleW
4/23/2021 10:16 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64; rv:78.0)
4/23/2021 10:16 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64; rv:78.0)
4/23/2021 10:15 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64; rv:78.0)
4/23/2021 10:15 PM				The information you entered was not recognized.	Mozilla/5.0 (X11; Linux x86_64; rv:78.0)


```
1 <pyIDPIdentifier>http://www.okta.com/ /</pyIDPIdentifier>
2 <pyIsSigningDisabled>>false</pyIsSigningDisabled>
3 <pySPIdentifier>https://kms.deere.com/ </pySPIdentifier>
4 <pyIDPSigningCertificateExpiry>Tue Aug 03 02:41:26 CDT 2027</pyIDPSigningCertificateExpiry>
5 <pyOriginalDecryptionPasswordRUF />
6 <pyIDPCertificateToVerifySignature />
7 <pyOriginalSignaturePassword>
pyOriginalSignaturePassword>
8 <pyOriginalSignaturePasswordRUF />
9 <pyOriginalDecryptionUser>prod</pyOriginalDecryptionUser>
10 <pyOriginalDecryptionPassword>
pyOriginalDecryptionPassword>
11 <pyIDPSigningCertificateAlias>EMAILADDRESS=info@okta.com, CN=johndeerecustomer,
OU=SSOProvider, O=Okta, L=San Francisco, ST=California, C=US</pyIDPSigningCertificateAlias>
12 <pyOriginalDecryptionPasswordSource>Direct</pyOriginalDecryptionPasswordSource>
13 <pyIsImportMetadataSuccess>true</pyIsImportMetadataSuccess>
14 <pyTemporaryObject />
15 <pzInsKey />
16 <pyOriginalIDPSigningCertificateAlias>emailaddress=info@okta.com, cn=
ou=ssoprovder, o=okta, l=san francisco, st=california, c=us</
pyOriginalIDPSigningCertificateAlias>
17 <pySingleSignOnServiceURL>https://signin.johndeere.com/app/johndeere_saml_1/
</pySingleSignOnServiceURL>
18 <pySPSigningCertificateExpiry>Mon Sep 10 07:39:38 CDT 2029</pySPSigningCertificateExpiry>
19 <pyDecryptionPassword> </pyDecryptionPassword>
```

CASE IH + NEW HOLLAND



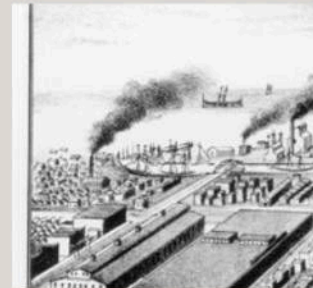
PRODUCTS ▾

COMPANY ▾

CONNECT WITH US ▾

SERVICE & SUPPORT ▾

TOOLS & RESOURCES ▾



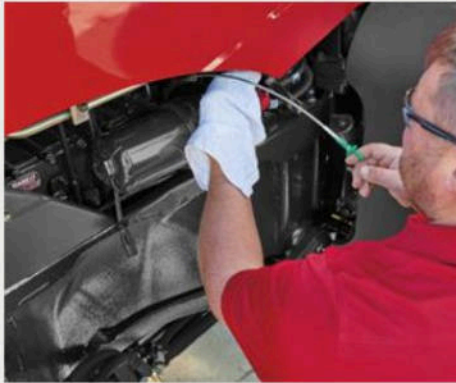


<https://www.caseih.com/northamerica/en-us/products/tractors/afs-connect-magnum-series>

MAINTENANCE & UPTIME

Remote Support and Resources to Maximize Productivity

Whether you are three miles or 300 miles away, you can get the support you need from your knowledgeable Case IH dealer. New features allow them to better support you and your fleet remotely — and in real time.



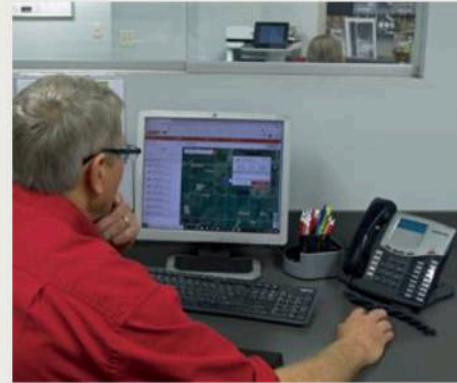
Ground-level Fluid Checks

Ground-level fluid checks from a single location near the cab entry help you to stay on top of maintenance as quickly as possible.



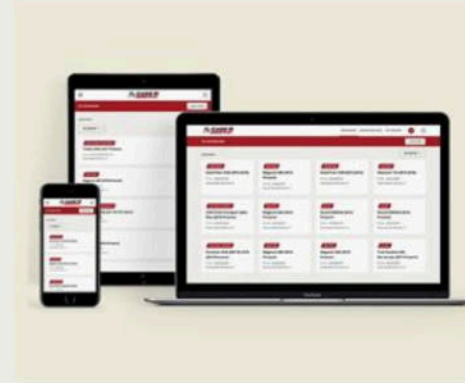
600-hour Oil Change Intervals

Stay in the field longer with Industry leading oil service change intervals.



Remote Dealer Support

From setting up your new AFS Connect Magnum tractor to diagnosing machine codes, your dealer is just a call away to provide you remote support.



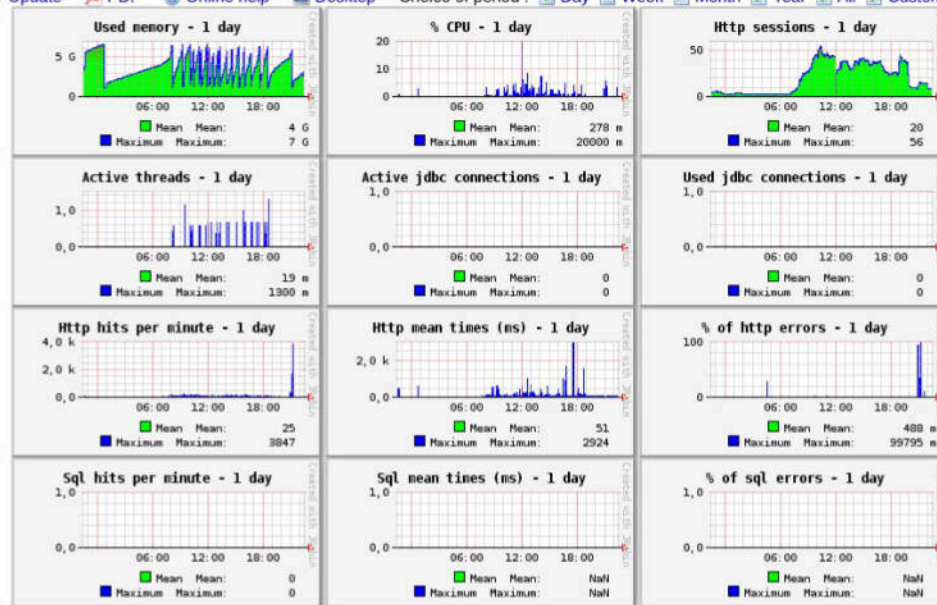
Information at Your Fingertips

MyCaseIH.com is your all-new destination for product support, including free operators manual downloads, AFS training videos, how-to tutorials and maintenance tips.



Statistics of **JavaMelody** monitoring taken at 4/28/21 10:24 PM on _MIBHFCATC03 (nh360)

Update PDF Online help Desktop Choice of period : Day Week Month Year All Customized



Other charts

Statistics http - 1 day

Request	% of cumulative time	Hits	Mean time (ms)	Max time (ms)	Standard deviation	% of cumulative cpu time	Mean cpu time (ms)	% of system error	Mean size (Kb)
http global	100	37,739	47	84,828	676	100	15	26.13	
http warning	8	142	1,095	13,111	2,183	5	210	0.00	
http severe	28	115	4,451	84,828	10,788	4	240	0.00	2

28 hits/min on 572 requests Details

Request	% of cumulative time	Hits	Mean time (ms)	Max time (ms)	Standard deviation	% of cumulative cpu time	Mean cpu time (ms)	% of system error	Mean size (Kb)
/ajax/repository/standardResource/sendMinutesPreviewEmail ajax POST	14	7	36,895	84,828	26,553	1	847	0.00	
/ajax/repository/directMessage/refresh ajax POST	10	3,355	56	4,063	117	21	36	0.00	
/pages/portal GET	4	206	410	3,932	413	9	275	0.00	
/pages/user/userManagerEdit POST	4	64	1,119	4,383	2,366	1	142	0.00	
/ajax/repository/strategicPlanResource/loadItem ajax POST	3	191	331	1,227	260	6	186	0.00	
/ajax/dynamicSelect/responsibleSelectUser ajax GET	3	450	135	289	205	6	78	0.00	
/pages/pm/cc/receipt/pmCcMediaReceiptManagerEdit POST	1	22	1,445	7,678	4,182	0	112	0.00	
/pages/account/tempPassword POST	1	14	2,265	7,129	3,088	0	66	0.00	
/pages/actionPlan/actionPlanItemEdit POST	1	10	3,094	5,888	4,419	0	97	0.00	
/ajax/118n/resources/1 GET	1	55	497	3,220	683	0	54	0.00	5
/ajax/repository/standardResource/getModuleTotal ajax POST	1	26	1,030	10,138	2,212	1	325	0.00	
/css/ace.min.css GET	1	307	85	2,839	327	0	4	0.00	

 Details session 6

Session id	Last access	Age	Expiration	Number of attributes	Serializable	Serializable size (b)	IP address	Country	Browser	OS	Invalidate
	00:00:18	01:10:17	5/4/21 8:49 AM	5	no	-1	200.218.138.137				

Attributes

Name	Type	Serializable	Serializable size (b)	Content
javamelody.sessionActivation	net.bull.javamelody.SessionListener	yes	75	SessionListener[sessionCount=11]
javamelody.userAgent	java.lang.String	yes	137	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML like Gecko) Chrome/90.0.4430.72 Safari/537.36 Edg/90.0.818.41
javamelody.remoteAddr	java.lang.String	yes	22	200.218.138.137
javamelody.country	java.lang.String	yes	9	BR
DirectMessageHandler-Object	br.com.usecase.cnh.view.directmessage.DirectMessageHandler	no	-1	br.com.usecase.cnh.view.directmessage.DirectMessageHandler@1911

25	0:21	proftpd: (accepting connections)
59	24:36	/usr/bin/java -Djava.util.logging.config.file=/home/fcatomcat/tomcat8/conf/logging.properties - Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Xms4096m -Xmx12288m -XX:NewSize=4096m - XX:MaxNewSize=12288m -XX:PermSize=4096m -XX:MaxPermSize=12288m -Djdk.tls.ephemeralDHKeySize=2048 - Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Djava.library.path=/usr/lib/x86_64-linux-gnu - classpath /home/fcatomcat/tomcat8/bin/bootstrap.jar:/home/fcatomcat/tomcat8/bin/tomcat-juli.jar - Dcatalina.base=/home/fcatomcat/tomcat8 -Dcatalina.home=/home/fcatomcat/tomcat8 - Djava.io.tmpdir=/home/fcatomcat/tomcat8/temp org.apache.catalina.startup.Bootstrap start
22	0:00	\ /bin/sh -c ps auxf

 Sessions

Session id	Last access	Age	Expiration	Number of attributes	S
	00:00:00	01:44:51	4/29/21 4:23 PM	5	
	00:00:05	03:19:12	4/29/21 4:22 PM	19	
	00:00:08	00:48:36	4/29/21 4:22 PM	5	
	00:00:18	00:36:00	4/29/21 4:22 PM	18	
	00:00:25	03:20:25	4/29/21 4:22 PM	5	
	00:00:43	01:51:47	4/29/21 4:22 PM	16	
	00:00:52	00:17:08	4/29/21 4:22 PM	18	
	00:01:01	01:16:31	4/29/21 4:22 PM	18	
	00:01:14	03:21:06	4/29/21 4:21 PM	5	
	00:01:32	01:31:41	4/29/21 4:21 PM	18	
	00:01:46	01:42:38	4/29/21 4:21 PM	18	
	00:02:14	01:45:48	4/29/21 4:20 PM	22	
	00:02:25	00:17:29	4/29/21 4:20 PM	16	
	00:03:02	03:13:26	4/29/21 4:20 PM	18	
	00:03:17	00:48:36	4/29/21 4:19 PM	21	
	00:03:22	03:19:58	4/29/21 4:19 PM	18	
	00:03:27	01:51:54	4/29/21 4:19 PM	5	
	00:03:36	01:48:22	4/29/21 4:19 PM	16	
	00:03:37	01:49:36	4/29/21 4:19 PM	6	
	00:04:08	03:19:50	4/29/21 4:18 PM	17	
	00:04:17	03:19:17	4/29/21 4:18 PM	5	



Details session

Session id	Last access	Age	Expiration	Number of attributes	Serializable
	00:01:43	05:03:01	4/28/21 10:41 PM	20	no

Attributes

Name	Type	Serializable	Serializable size (b)	
topMenuButton_0	java.lang.String	yes	11	null
__uc_DOMAIN_ID	java.lang.Long	yes	82	1
GDSP_STEP_PREFIX_baseReportTasks	java.lang.Integer	yes	81	6
ScopeMenuHandler	br.com.usecase.cnh.view.menu.scope.ScopeMenuHandler	no	-1	br.com.usecase.cnh.view.menu.scope.ScopeMenuHandler@
javal melody.userAgent	java.lang.String	yes	121	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/90.0.4430.9
GDSP_STEP_PREFIX_dealershipSelectorQPv	java.lang.Integer	yes	81	1
javal melody.remoteAddr	java.lang.String	yes	2	
javal melody.country	java.lang.String	yes	9	BR
topMenuButton_3	java.lang.String	yes	11	null
GDSP_TOTAL_PREFIX_baseReportTasks	java.lang.Integer	yes	81	4
topMenuButton_2	java.lang.String	yes	11	null
DirectMessageHandler-Object	br.com.usecase.cnh.view.directmessage.DirectMessageHandler	no	-1	br.com.usecase.cnh.view.directmessage.DirectMessageHandler@4dc7e08b
__uc_AUTHENTICATED_USER	br.com.usecase.fwcnh.common.security.AuthenticatedUser	yes	629	AuthenticatedUser [id= name=
topMenuButton_1	java.lang.String	yes	11	null
javal melody.sessionActivation	net.bull.javamelody.SessionListener	yes	75	SessionListener[sessionCount=15]
				<li class=" minutes minutesList minutes report minutesGeneralReport actionPlan actionPlanList acti

Error

WARN [org.hibernate.hql.internal.ast.QueryTranslatorImpl] HH-H000104: firstResult/maxResults specified with collection fetch; applying in memory!



Current requests

None



System information

Host:	[REDACTED]	
Java memory used:	1,099 Mb / 8,192 Mb	
Nb of http sessions:	9	
Nb of active threads (current http requests):	0	
Nb of active jdbc connections:	0	
Nb of used jdbc connections (opened if no datasource):	0	
System load	0.67	
% System CPU	1.48	



Execute the garbage collector



Generate a heap dump



View memory histogram



Invalidate http sessions



View http sessions



View deployment descriptor



MBeans



View OS processes



JNDI tree

[Details](#)



Threads

Threads on [REDACTED] Number = 54, Maximum = 54, Total started = 679 [Details](#)

Last collect time: 68 ms

Display time: 06 ms



Execute the garbage collector



Generate a heap dump



View deployment descriptor



View memory histogram



MBeans



View OS processes



Invalidate http sessions



JNDI tree

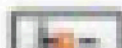


View http sessions

 Execute the garbage collector  Generate a heap dump  View memory histogram  Invalidate http sessions  View http sessions
 View deployment descriptor  MBeans  View OS processes  JNDI tree



Invalidate http sessions



JNDI tree

williecade@gmail.com

From: williecade@gmail.com
Sent: Tuesday, April 13, 2021 1:35 PM
To: MayJohnC@johndeere.com
Cc: 'Sick.Codes'; [REDACTED]@johndeere.com; [REDACTED]@johndeere.com
Subject: Data Security Disruption
Attachments: John Deere GDPR Data Security and Privacy Risk Identified.docx

Mr. May,

Please accept this email in the helpful spirit intended. I am the grandson of Theo Brown who served on the Board of Directors of John Deere for 30 years.

I along with another researcher, Sick Codes, have discovered a vulnerability that exposes your customers Personnel Identifying Information and other issues. We believe this is a significant and urgent matter that respectfully deserves your attention. I have attached a summary of the issue to this email. Given well established notification protocol I am not allowed to directly share the detailed vulnerability reports. Rest assured that Mr. [REDACTED] (@JohnDeere.com) and the API Support team has been sent both reports.

I will be the point of contact going forward.

Thank you.

Willie Cade

Cell: [REDACTED]

"A nation that isn't broken but simply not finished" Amanda Gorman, 20-Jan-2021



COMPLIANCE

CNH Industrial believes that operating in a socially responsible and ethical manner and in compliance with the laws of those countries in which we operate is fundamental to our long-term success. This means, among other things, that we adopt fair employment practices, protect safety in the workplace, support and foster environmental consciousness and fully comply with applicable laws.

CNH Industrial Compliance Helpline

The Board of Directors has established a procedure to ensure that the Company's employees and third parties have the possibility to report alleged irregularities of a general, operational and financial nature with the Company.

The Company's compliance helpline is managed by an independent third party and is available 24 hours per day/seven days per week/365 days per year. Reports may be submitted through a dedicated web portal (www.cnhindustrialcompliancehelpline.com), by phone (to a call center managed by a third party), or to a Company representative. Where legally permissible, reports may be submitted on an anonymous basis. In addition, where legally required, the nature of the reports may be limited to certain subject matters. The Company investigates reports submitted and, in appropriate cases, implements corrective actions.

If you become aware of a circumstance or action that violates, or appears to violate, CNH Industrial's Code of Conduct, company policies or applicable law, you can seek guidance, or report a violation, by using the CNH Industrial Compliance Helpline, available at www.cnhindustrialcompliancehelpline.com.

Related Links

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Our Compliance Helpline

This tool is available to all our stakeholders, such as employees, customers, dealers and suppliers, for asking a question or reporting possible violations of our Code of Conduct, other policies or applicable laws. You may choose to submit a report by providing contact information, which will be held in the strictest of confidence.

[Read More...](#)



ASK A QUESTION

Click here if you have a question about our Code of Conduct



RAISE A CONCERN

Click here to raise a concern about unethical behavior in the workplace



FOLLOW UP

Click here to follow up on a previously submitted question or concern



632

8 comments



Like



Comment



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Send

Reactions



Comments

Most relevant



Sick Codes

1w ...

Security Researcher, Freelance Automation Speci...

Can someone at CNH get in touch regarding a possible security vulnerability?

Like Reply • 1 reply



Sick Codes

5d ...

Security Researcher, Freelance Automation...

Still waiting, multiple phone calls and emails, please get in touch.

Like Reply



Leave your thoughts here...

@ Post



Home



My Network



Post



Notifications



Jobs



SHIPPING & RECEIVING ▶
Use phone @ gate for access

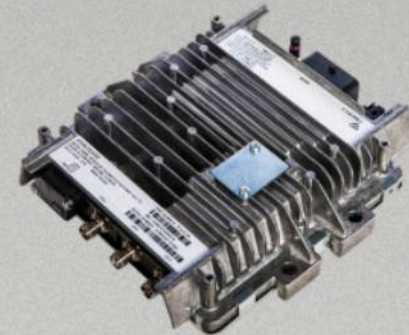
SECURITY OFFICE ▶

HUMAN RESOURCES ▶

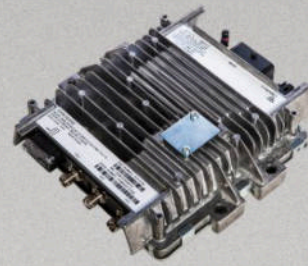
MTG 4G LTE



The Modular Telematics Gateway integrates GNSS, mobile connectivity and on-board machine communications via CAN, Ethernet, and RS-232. It's ideally suited for advanced telematics applications such as machine health monitoring, advanced logistics, and remote diagnostics.



The Modular Telematics Gateway integrates GNSS, mobile connectivity and on-board machine communications via CAN, Ethernet, and RS-232. It's ideally suited for advanced telematics applications such as machine health monitoring, advanced logistics, and remote diagnostics.



Storage: 8GB
onboard flash



Designed and tested to
extreme environments



Supports LTE with fallback to 3G
and 2G cellular communication



Supports 12V and
24V systems



Open source created with
Yocto Project® software*



Supports multiple SIM options:
- Internal soldered SIM IC
- External accessible Micro-SIM
- Combination of internal and
external SIM

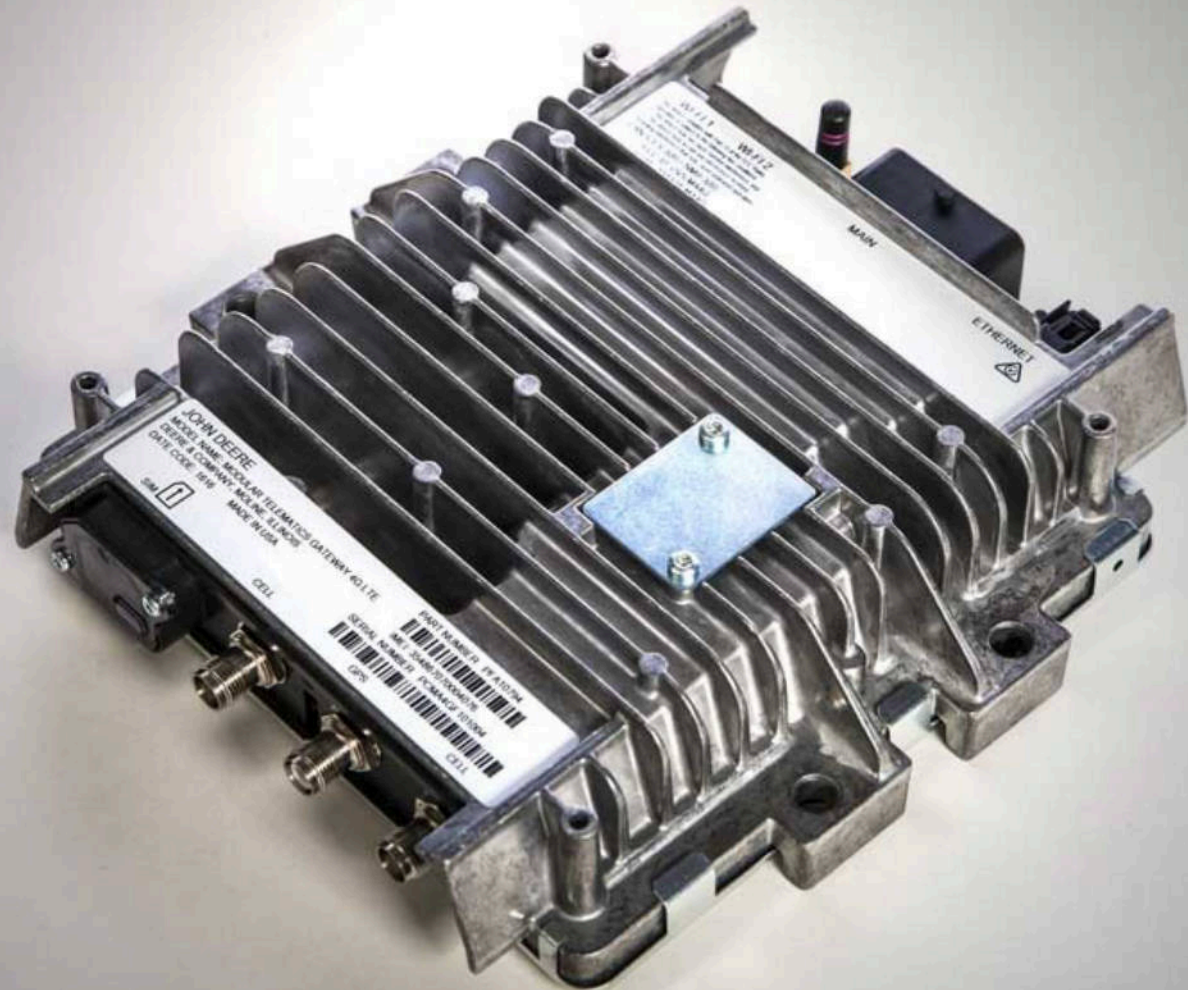


Wireless and Bluetooth®
communication



Certified in over
70 countries

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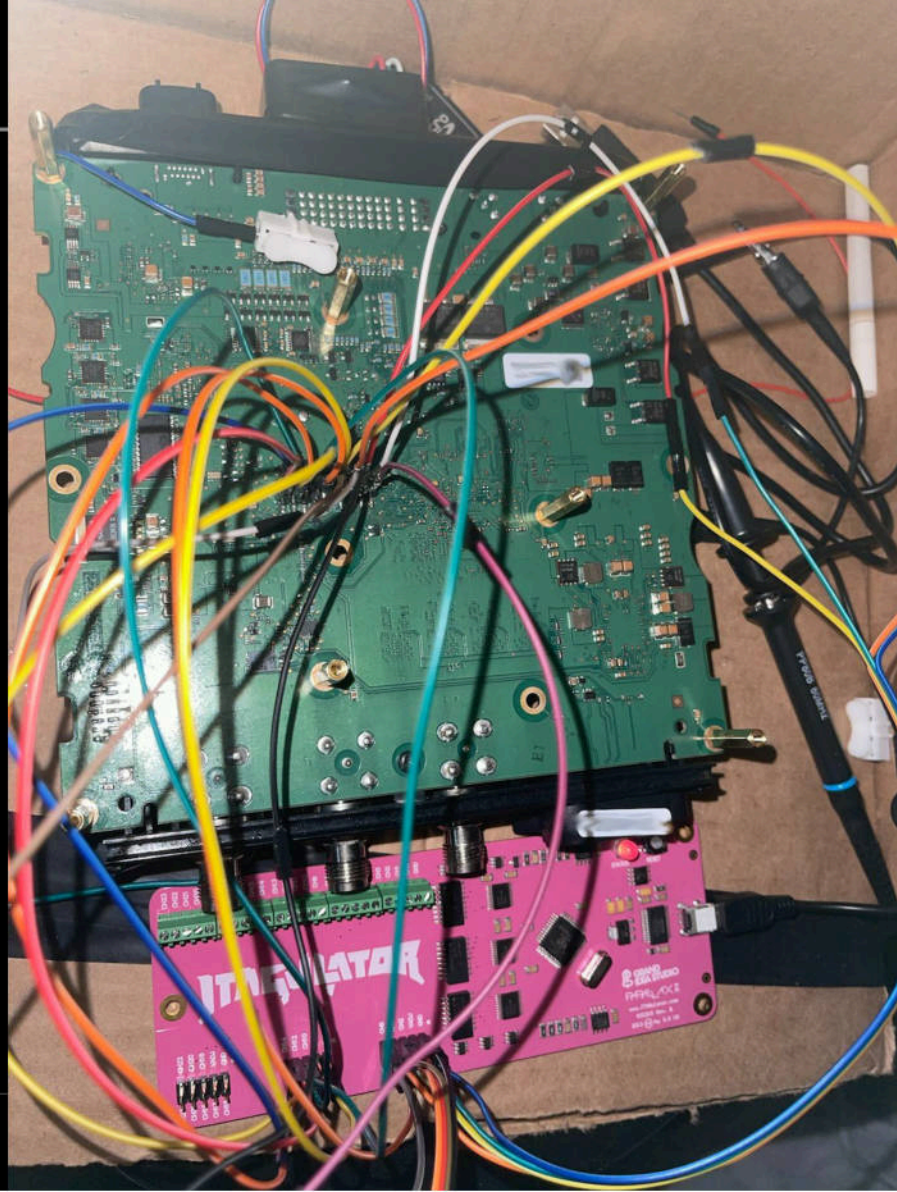


JOHN DEERE
MODEL NAME: MCX44N TELEMATICS GATEWAY V01.1E
DATE CODE: 1019
MADE IN USA

CELL
1001 10000000 10000000
SERIAL NUMBER: 10000000000000000000
10000000000000000000
10000000000000000000

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WIFI 99
WIFI 100

ETHERNET



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Copies of those licenses are included in the Third Party Notices.

You may obtain the complete Corresponding Source Code from us for a period of three years after our last shipment of the Software by sending a request letter to:

Deere Open Source Compliance Team
P.O. Box 1202
Moline, IL 61266-1202 USA

Please include "source for John Deere MTG 4G LTE" and the version number of the software in the request letter. This offer is valid to anyone in receipt of this information.

AE77568,000020D -19-23JUN16-1/1

- [https://
www.qualcomm.
com/company/
product-
security/
bulletins](https://www.qualcomm.com/company/product-security/bulletins)



Public ID	Security Rating	CVSS Rating	Technology Area	Date Reported
<u>CVE-2020-11307</u>	Critical	Critical	Data Network Stack & Connectivity	Internal
<u>CVE-2021-1886</u>	Critical	High	HLOS	Internal
<u>CVE-2021-1888</u>	Critical	High	HLOS	Internal
<u>CVE-2021-1889</u>	Critical	High	HLOS	Internal
<u>CVE-2021-1890</u>	Critical	High	HLOS	Internal
<u>CVE-2021-1887</u>	High	High	WLAN Firmware	08/06/2020
<u>CVE-2021-1938</u>	High	High	WLAN Firmware	11/20/2020
<u>CVE-2021-1953</u>	High	High	WLAN Firmware	12/16/2020

This table lists moderate security vulnerabilities. OEMs have been notified and encouraged to patch these issues.

Thank you.

- Sick Codes <https://twitter.com/sickcodes>
- wabaf3t <https://twitter.com/wabafet1>
- D0rkerDevil <https://twitter.com/D0rkerDevil>
- johnjhacking <https://twitter.com/johnjhacking>
- rej_ex https://twitter.com/rej_ex
- w0rmer <https://twitter.com/0x686967>
- ChiefCoolArrow <https://twitter.com/ChiefCoolArrow>
- Kelly <https://twitter.com/kaoudis>
- Willie Cade <https://twitter.com/WillieCade7>
- Kevin Kenney <https://twitter.com/GrassrootsKK>